

Computational Physics: Materials Science

Exercise 3

Wave packet dynamics in a double-well potential

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Wave Packet Dynamics in a Double-Well Potential

This exercise simulates the quantum mechanical time evolution of a Gaussian wave packet in a symmetric quartic double-well potential using the **split-operator method**. Quantum dynamics are compared against classical trajectories; tunnelling, energy conservation, and mass dependence are studied.

Implementation

Goal and Physical Setup

The particle moves in a double-well potential with minima at $x = \pm d$ and barrier V_b :

$$V(x) = V_b \frac{(x^2 - d^2)^2}{d^4}. \quad (1)$$

All calculations use **Hartree atomic units** internally ($\hbar = 1$, $m_e = 1$) with parameters $V_b = 0.15 E_h$ (≈ 4.08 eV), $d = 2.0 a_0$ (≈ 1.06). The zero-momentum Gaussian initial state is

$$\psi(x, 0) = \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^{1/2} \exp\left(-\frac{(x - x_0)^2}{4\sigma^2} \right), \quad x_0 = -d, \sigma = 0.5 a_0 (\approx 0.26). \quad (2)$$

With $m_H = 1836.15 m_e$ the total initial energy is $\langle E \rangle_H \approx 1.08$ eV $< V_b \approx 4.08$ eV (**sub-barrier**, $\langle E \rangle/V_b = 0.264$): the classical particle ($E_{cl} = 0$) cannot cross while the quantum particle tunnels. All axis labels are in physical units (eV, Å, fs). An AI assistant (Claude Sonnet 4.6) assisted with coding; all physical interpretation is independent.

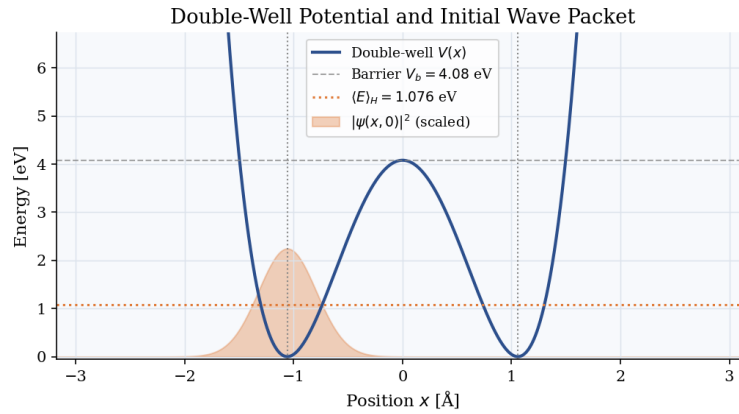


Figure 1: Double-well $V(x)$ (blue), barrier $V_b \approx 4.08$ eV (dashed), $\langle E \rangle_H \approx 1.08$ eV (dotted orange), and initial $|\psi|^2$ (orange, scaled). Sub-barrier: $\langle E \rangle/V_b = 0.264$.

Numerical Method: Split-Operator Propagation

The TDSE is integrated via second-order Trotter splitting (split-operator method):

$$\psi(x, t + \Delta t) = e^{-iV\Delta t/2\hbar} \mathcal{F}^{-1} \left[e^{-i\hbar k^2 \Delta t/(2m)} \mathcal{F} \left[e^{-iV\Delta t/2\hbar} \psi \right] \right], \quad (3)$$

where $\mathcal{F}$, $\mathcal{F}^{-1}$ are the FFT and its inverse. Grid: $x \in [-6, 6] a_0$, $N = 1024$. The maximum stable timestep is $\Delta t_{\max} \approx 3.89$ as; the fine timestep used is $\Delta t \approx 2.42$ as ($\Delta t/\Delta t_{\max} = 0.62$, within stability limit), with 500,000 steps giving a total of ≈ 1210 fs. The classical particle uses Velocity-Verlet integration.



Analysis

1. Quantum vs. Classical Trajectory

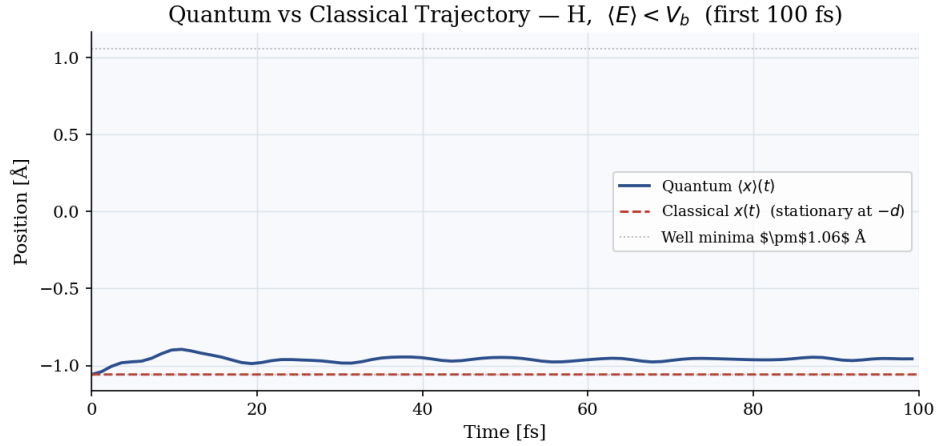


Figure 2: Quantum $\langle x \rangle(t)$ (blue) vs. classical $x(t)$ (red dashed, stationary at $x = -d \approx -1.06$), first 100 fs. The classical particle stays fixed; $\langle x \rangle$ drifts toward zero as probability tunnels through the classically forbidden barrier.

The classical particle remains fixed at $x_0 = -d$ ($E_{\text{cl}} = 0 < V_b$). The quantum $\langle x \rangle(t)$ drifts toward zero within 100 fs, demonstrating probability leaking through the barrier via quantum tunnelling.

2. Well Populations

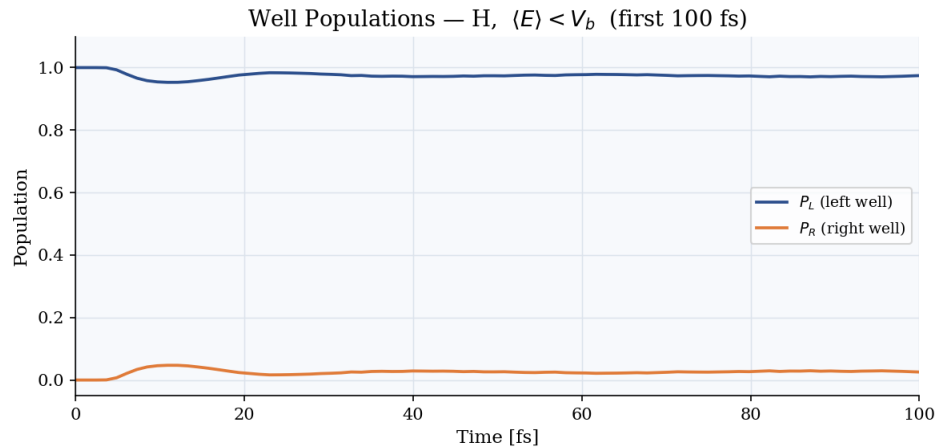


Figure 3: P_L (blue) and P_R (orange) vs. time, first 100 fs, where $P_{L,R}(t) = \int_{\mp\infty}^0 |\psi|^2 dx$. From $P_L = 1$ at $t = 0$, probability gradually transfers to the right well; the classical particle stays at $P_L = 1$ permanently.

3. Energy Conservation and Timestep Comparison

$$\langle T \rangle = \sum_k |\hat{\psi}_k|^2 T_k \cdot \Delta x / N \text{ (Parseval)} \text{ and } \langle V \rangle = \int V |\psi|^2 dx.$$



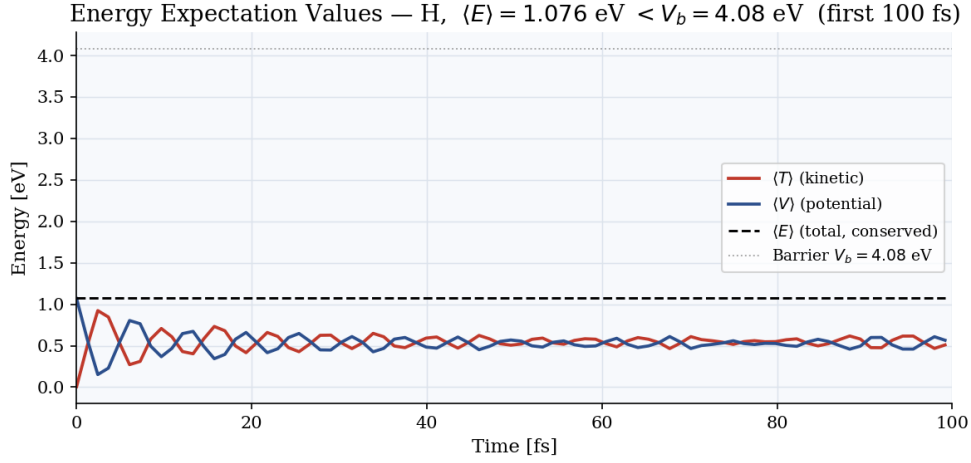


Figure 4: Energy expectation values ($\Delta t \approx 2.42$ as, first 100 fs). $\langle T \rangle$ (red) and $\langle V \rangle$ (blue) oscillate in anti-phase; $\langle E \rangle \approx 1.076$ eV (black dashed) is conserved to $< 4 \times 10^{-5} \%$. Grey dotted: $V_b \approx 4.08$ eV.

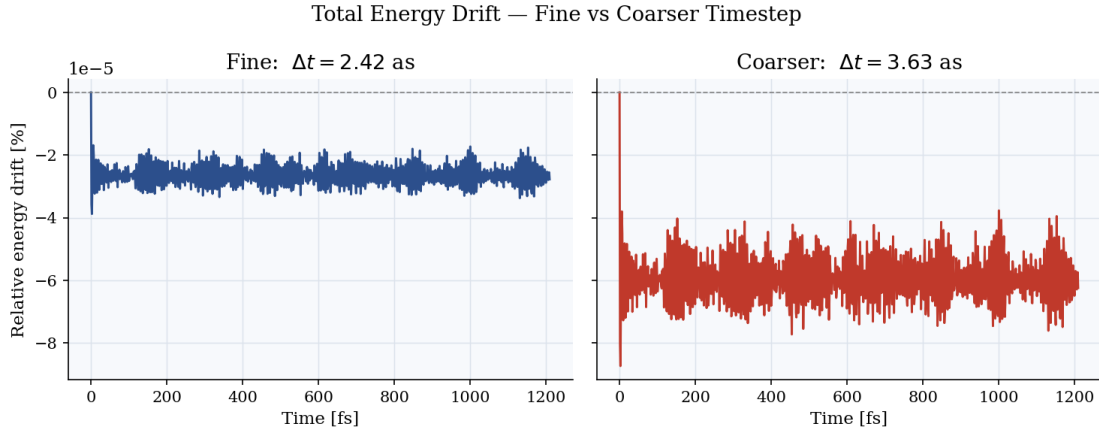


Figure 5: Relative energy drift $(\langle E \rangle(t) - \langle E \rangle(0)) / |\langle E \rangle(0)|$ for the fine ($\Delta t \approx 2.42$ as, blue) and coarser ($\Delta t \approx 3.63$ as, red) timesteps. Maximum drifts: $3.9 \times 10^{-5} \%$ and $8.7 \times 10^{-5} \%$ ($\approx 2 \times$ larger), consistent with $\mathcal{O}(\Delta t^2)$ global Trotter error.

The split-operator method is unitary to machine precision, but Trotter splitting accumulates $\mathcal{O}(\Delta t^2)$ global phase error. Both timesteps are within $\Delta t_{\max}$; the coarser choice shows proportionally larger drift.

4. Above-Barrier Regime: $\langle E \rangle > V_b$

A plane-wave kick $\psi_{\text{hi}} = \psi_0 e^{ik_0 x}$, $k_0 = \sqrt{2m_H \cdot 2V_b}/\hbar$, gives $\langle E \rangle \approx 9.23$ eV $\approx 2.26 V_b > V_b$.



Above-Barrier Regime: $\langle E \rangle = 9.23 \text{ eV} > V_b = 4.08 \text{ eV}$ — Quantum vs Classical (H)

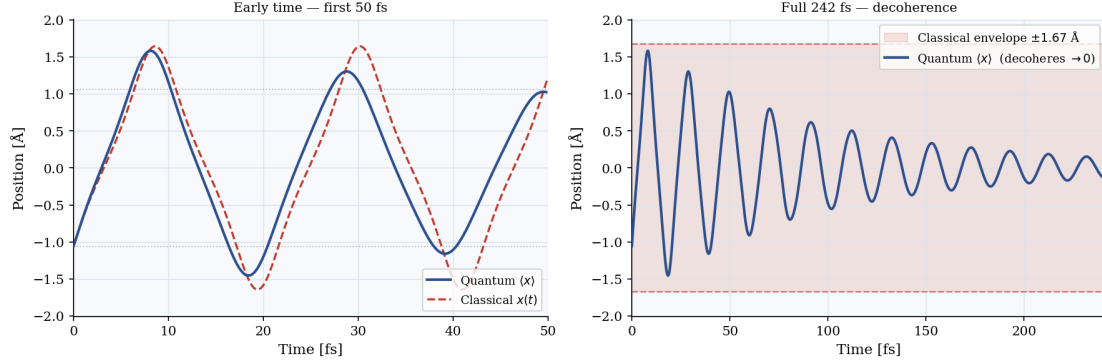


Figure 6: **Left:** first 50 fs — quantum (blue) and classical (red dashed) both oscillate freely above the barrier. **Right:** quantum $\langle x \rangle$ decoheres to zero as the packet spreads symmetrically; classical amplitude ($\pm x_{tp}$, shaded) persists indefinitely.

Above the barrier the classical particle oscillates freely. The quantum $\langle x \rangle$ rapidly decoheres to zero as different momentum components traverse the potential at different speeds, producing destructive interference — a purely quantum effect.

5. Wavefunction Snapshots

Probability Density Snapshots — H (sub-barrier, first 100 fs)

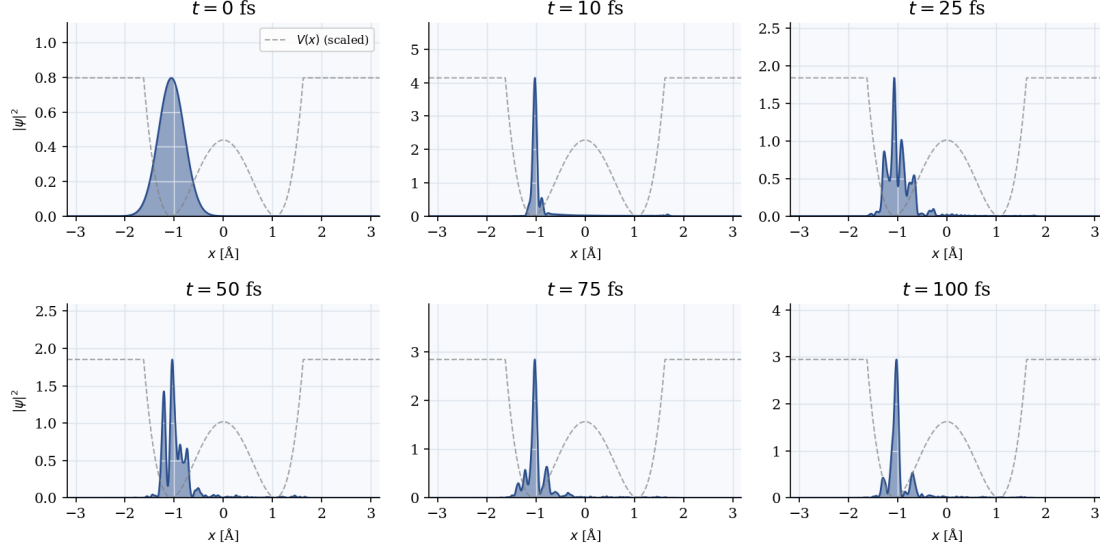


Figure 7: $|\psi(x,t)|^2$ at $t = 0, 10, 25, 50, 75, 100 \text{ fs}$ (dashed gray: scaled $V(x)$). A secondary right-well peak grows progressively, confirming quantum tunnelling on a femtosecond timescale.

At $t = 0$ the packet is localised in the left well. By $\sim 10 \text{ fs}$ a small secondary peak appears in the right well and grows steadily through 100 fs, providing a clear spatial picture of the tunnelling process.



6. Mass Comparison: Hydrogen vs. Carbon

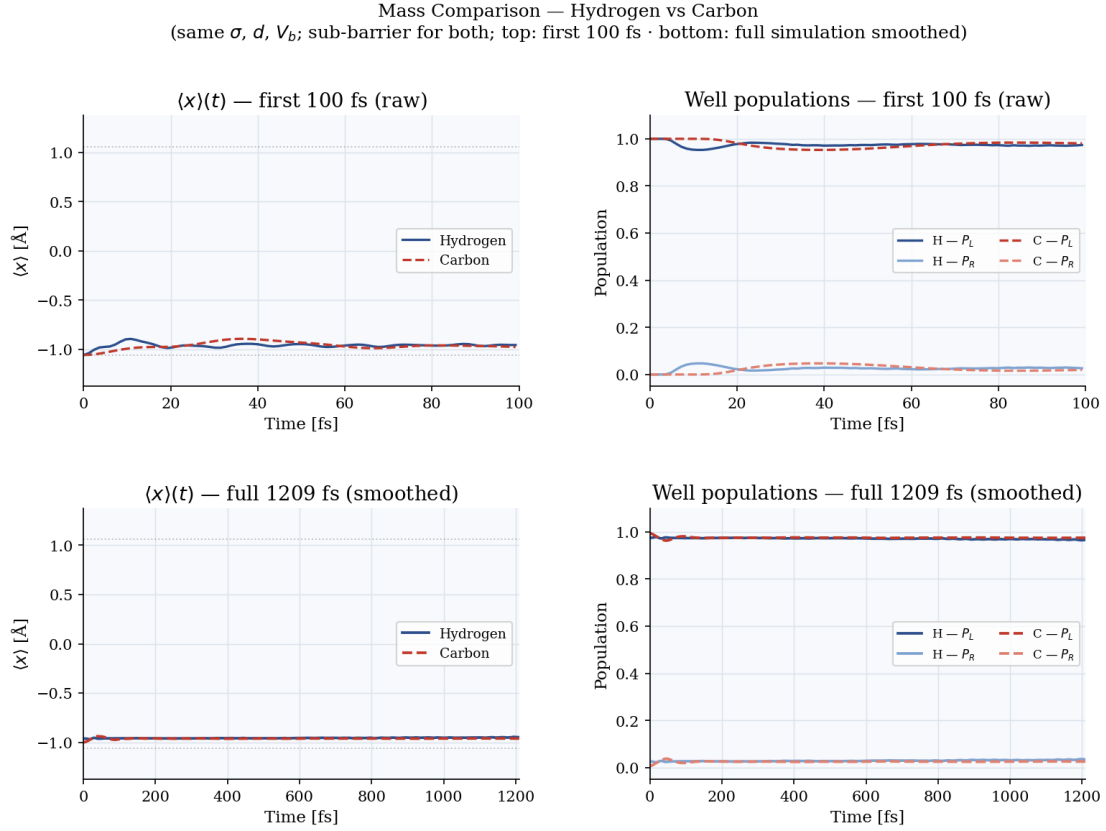


Figure 8: $\langle x \rangle(t)$ (left) and well populations (right) for H (blue/light-blue) and C (red/orange), both sub-barrier ($\langle E \rangle / V_b < 1$). **Top:** first 100 fs — H spreads noticeably faster than C. **Bottom:** full 1210 fs (smoothed ~ 50 fs) — H equilibrates while C remains predominantly in the left well.

Hydrogen spreads faster than carbon within 100 fs. The WKB tunnelling rate is exponentially suppressed with mass:

$$\Gamma \propto \exp\left(-\frac{2}{\hbar} \int_{x_1}^{x_2} \sqrt{2m(V(x) - E)} dx\right). \quad (4)$$

Since $m_C \approx 12 m_H$, carbon's WKB exponent is $\sqrt{12} \approx 3.5 \times$ larger, confirming tunnelling as a light-particle phenomenon.

7. Parameter Exploration: Barrier Geometry

Simulations were repeated for a 2×2 grid: $d \in \{0.79, 1.59\} \text{ Å}$, $V_b \in \{0.82, 4.08\} \text{ eV}$.



Potential Shapes — 2×2 Parameter Combinations

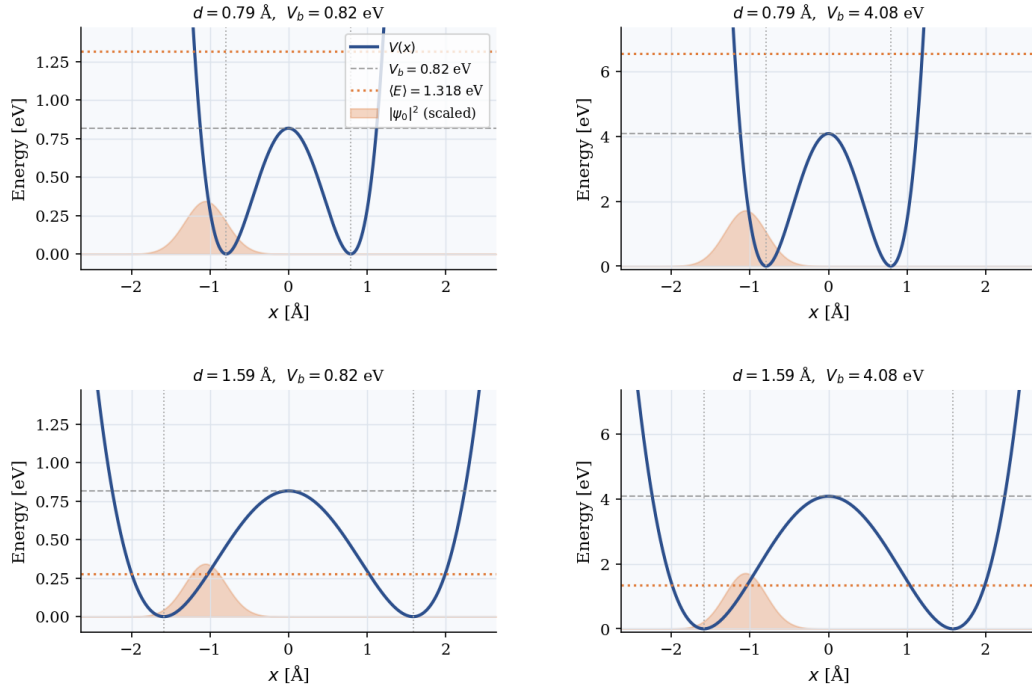


Figure 9: Potential shapes for all four parameter combinations. Narrow well ($d \approx 0.79$, top row): above-barrier ($\langle E \rangle / V_b \approx 1.61$) — the Gaussian sits above the barrier regardless of V_b . Wide well ($d \approx 1.59$, bottom row): sub-barrier ($\langle E \rangle / V_b \approx 0.33$); the lower barrier (left) allows faster spreading.



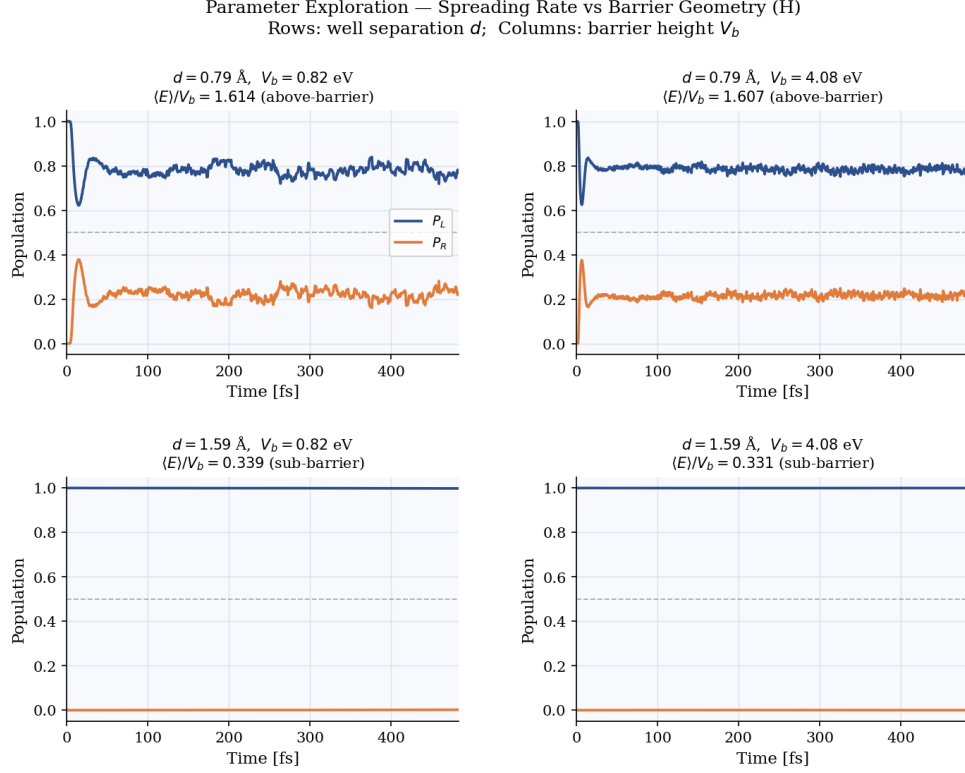


Figure 10: Well populations P_L (blue) and P_R (orange). Above-barrier cases (top row, $d \approx 0.79$): both wells fill immediately. Sub-barrier cases (bottom row, $d \approx 1.59$): the lower barrier ($V_b \approx 0.82$ eV, left) allows noticeably faster spreading than the higher barrier ($V_b \approx 4.08$ eV, right).

A narrower d samples steeper walls, pushing $\langle E \rangle > V_b$ regardless of V_b . For the wide-well sub-barrier case, reducing V_b from 4.08 eV to 0.82 eV produces clearly faster spreading — directly demonstrating the exponential WKB dependence on barrier height.

Summary

- **Split-operator method** in Hartree atomic units ($\hbar = 1$, $m_e = 1$). Fine timestep $\Delta t \approx 2.42$ as: energy conserved to $< 4 \times 10^{-5} \%$; coarser $\Delta t \approx 3.63$ as gives $\approx 2 \times$ more drift ($\mathcal{O}(\Delta t^2)$ Trotter error).
- **Sub-barrier regime** ($\langle E \rangle \approx 1.08$ eV $< V_b \approx 4.08$ eV, ratio 0.264): quantum wave packet tunnels within ~ 100 fs; classical particle remains stationary at the left-well minimum.
- **Above-barrier regime** ($\langle E \rangle \approx 9.23$ eV $\approx 2.26 V_b$): classical particle oscillates freely; quantum $\langle x \rangle$ decoheres to zero rapidly via destructive interference.
- **Wavefunction snapshots** at $t = 0, 10, 25, 50, 75, 100$ fs confirm progressive femtosecond tunnelling from left into right well.
- **Mass comparison**: H ($m_H = 1836 m_e$) tunnels readily; C ($m_C \approx 12 m_H$) is exponentially WKB-suppressed — tunnelling is a light-particle phenomenon.
- **Parameter exploration**: wider d keeps system sub-barrier; narrower d pushes $\langle E \rangle > V_b$. Reducing V_b from 4.08 eV to 0.82 eV produces visibly faster spreading, demonstrating the exponential WKB barrier-height dependence.

